

18 A

Standing wave on a rope has nodes at both ends.

5

DUNMAN HIGH SCHOOL
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$$L = n \frac{\lambda}{2} = 1.0$$

For wavelength 0.4 m, this corresponds to the 5th harmonic (n = 5) mode, thus A is correct.

B is wrong as there are 6 nodes for the 5th harmonic.

C is wrong as the fundamental wavelength is 2 m.

D is wrong as the midpoint of the rope may be moving (e.g. for fundamental mode, the midpoint is the antinode).